

OS LAB ASSIGNMENT – 2

1. Description of commands:

- 1) cut : remove sections from each line of a file. It has options:
 - -c : to cut a range of characters from each line of file.
 - -d : to cut fields (or words in .txt file) on basis of a delimiter(separator).
 - -f : to cut particular field (or word in .txt file).
- 2) uniq: It is used to filter out adjacent duplicate lines from a file. It often used after sorting a file. It has options:
 - -c : gives no. of occurrence of adjacent duplicate lines.
 - -d: gives lines which are duplicated.
 - -u :gives lines which have no duplicity in adjacent.
 - -l : facilitate ignorance of cases while comparing text lines.
- 3) chmod : It stands for 'change mode'. It used to change the access permissions for files or folders. To see which types of permissions (for a file/folder) , we can run command 'ls -l'.
- 4) du: It stands for 'disk usage'. It is used to estimate file space usage.
- 5) diff: It is used to compare files line by line. It displays the difference between two text files in a structured format.
- 6) history : It gives a sorted list of all the commands (since installation) ran so far.
- 7) last : It is used to display a list of previously logged in users and their login sessions.
- 8) sort: It is used sort (ascending/descending) lines of text files or standard input data. It has follow options:
 - -r : facilitates sorting in descending/reverse order.
 - -n : facilitates sorting in numerical order.
 - -u : suppress duplicate lines while sorting.
- 9) sed : It stands for 'Stream Editor'. It operates on a line-by-line basis, allowing users to perform various operations on text data such as search, replace, insertion, deletion and more. It has options:
 - s : to substitute old data with new one.
 - i : to insert a line/data in between.
 - a : to append.

10) grep : It is used for searching text patterns within files or standard input. It allows user to specify a pattern and searches for lines in files that matches that pattern.

11) pipe (|) : It is used to connect the output of one command to the input of another command . It allows chaining of multiple commands together, creating a pipeline where the output of one command becomes the input of next command.

2. Implementations of Commands: *Executed in ascending order:*

```
sachingola@DESKTOP-S4K7IL  ×  +  ∨
sachingola@DESKTOP-S4K7IU0:~$ man cut
sachingola@DESKTOP-S4K7IU0:~$ cut -c 1-6 thirdfile.txt
This i
It is
sachingola@DESKTOP-S4K7IU0:~$ cut -d ' ' -f 2 thirdfile.txt
is
is
sachingola@DESKTOP-S4K7IU0:~$ cat < anum.txt
-bash: anum.txt: No such file or directory
sachingola@DESKTOP-S4K7IU0:~$ cat > anum.txt
Hello Duniya.
^C
sachingola@DESKTOP-S4K7IU0:~$ cat < anum.txt
Hello Duniya.
sachingola@DESKTOP-S4K7IU0:~$ rm a*
sachingola@DESKTOP-S4K7IU0:~$ cat < anum.txt
-bash: anum.txt: No such file or directory
sachingola@DESKTOP-S4K7IU0:~$ cat > duplicate.txt
hello kohli
hello kohli
hello root
hello smith
hello root
hello root
^C
sachingola@DESKTOP-S4K7IU0:~$ uniq -c duplicate.txt
  2 hello kohli
  1 hello root
  1 hello smith
  2 hello root
sachingola@DESKTOP-S4K7IU0:~$ uniq -d duplicate.txt
hello kohli
hello root
sachingola@DESKTOP-S4K7IU0:~$ uniq -u duplicate.txt
hello root
hello smith
sachingola@DESKTOP-S4K7IU0:~$ cat >> duplicate.txt
HELLO ROOT
^C
sachingola@DESKTOP-S4K7IU0:~$ uniq -c duplicate.txt
  2 hello kohli
  1 hello root
  1 hello smith
  2 hello root
  1 HELLO ROOT
```

```
sachingola@DESKTOP-S4K7IUL x + v
sachingola@DESKTOP-S4K7IU0:~$ uniq -i duplicate.txt
hello kohli
hello root
hello smith
hello root
sachingola@DESKTOP-S4K7IU0:~$ uniq -i -c duplicate.txt
  2 hello kohli
  1 hello root
  1 hello smith
  3 hello root
sachingola@DESKTOP-S4K7IU0:~$ ls -l
total 24
-rw-r--r-- 1 sachingola sachingola  80 Jan 28 20:16 duplicate.txt
-rw-r--r-- 1 sachingola sachingola 133 Jan 28 19:15 file1.txt
-rw-r--r-- 1 sachingola sachingola 133 Jan 28 19:18 firstfile
drwxr-xr-x 2 sachingola sachingola 4096 Jan 28 19:35 seconddir
drwxr-xr-x 2 sachingola sachingola 4096 Jan 28 19:35 thirddir
-rw-r--r-- 1 sachingola sachingola  55 Jan 28 20:01 thirdfile.txt
sachingola@DESKTOP-S4K7IU0:~$ chmod u-w duplicate.txt
sachingola@DESKTOP-S4K7IU0:~$ ls -l
total 24
-r--r--r-- 1 sachingola sachingola  80 Jan 28 20:16 duplicate.txt
-rw-r--r-- 1 sachingola sachingola 133 Jan 28 19:15 file1.txt
-rw-r--r-- 1 sachingola sachingola 133 Jan 28 19:18 firstfile
drwxr-xr-x 2 sachingola sachingola 4096 Jan 28 19:35 seconddir
drwxr-xr-x 2 sachingola sachingola 4096 Jan 28 19:35 thirddir
-rw-r--r-- 1 sachingola sachingola  55 Jan 28 20:01 thirdfile.txt
sachingola@DESKTOP-S4K7IU0:~$ chmod 644 duplicate.txt
sachingola@DESKTOP-S4K7IU0:~$ ls -l
total 24
-rw-r--r-- 1 sachingola sachingola  80 Jan 28 20:16 duplicate.txt
-rw-r--r-- 1 sachingola sachingola 133 Jan 28 19:15 file1.txt
-rw-r--r-- 1 sachingola sachingola 133 Jan 28 19:18 firstfile
drwxr-xr-x 2 sachingola sachingola 4096 Jan 28 19:35 seconddir
drwxr-xr-x 2 sachingola sachingola 4096 Jan 28 19:35 thirddir
-rw-r--r-- 1 sachingola sachingola  55 Jan 28 20:01 thirdfile.txt
sachingola@DESKTOP-S4K7IU0:~$ chmod u-r duplicate.txt
sachingola@DESKTOP-S4K7IU0:~$ cat duplicate.txt
cat: duplicate.txt: Permission denied
sachingola@DESKTOP-S4K7IU0:~$ chmod u+r duplicate.txt
sachingola@DESKTOP-S4K7IU0:~$ cat duplicate.txt
hello kohli
hello kohli
hello root
hello smith
hello root
hello root
HELLO ROOT
```

```
sachingola@DESKTOP-S4K7IL x + v
sachingola@DESKTOP-S4K7IU0:~$ chmod u=r duplicate.txt
sachingola@DESKTOP-S4K7IU0:~$ ls -l
total 24
-r--r--r-- 1 sachingola sachingola 80 Jan 28 20:16 duplicate.txt
-rw-r--r-- 1 sachingola sachingola 133 Jan 28 19:15 file1.txt
-rw-r--r-- 1 sachingola sachingola 133 Jan 28 19:18 firstfile
drwxr-xr-x 2 sachingola sachingola 4096 Jan 28 19:35 seconddir
drwxr-xr-x 2 sachingola sachingola 4096 Jan 28 19:35 thirddir
-rw-r--r-- 1 sachingola sachingola 55 Jan 28 20:01 thirdfile.txt
sachingola@DESKTOP-S4K7IU0:~$ du duplicate.txt
4      duplicate.txt
sachingola@DESKTOP-S4K7IU0:~$ du /home/sachingola/seconddir
4      /home/sachingola/seconddir
sachingola@DESKTOP-S4K7IU0:~$ du -h /home/sachingola/seconddir
4.0K   /home/sachingola/seconddir
sachingola@DESKTOP-S4K7IU0:~$ du -s /home/sachingola/seconddir
4      /home/sachingola/seconddir
sachingola@DESKTOP-S4K7IU0:~$ du -c /home/sachingola/
4      /home/sachingola/thirddir
4      /home/sachingola/seconddir
48     /home/sachingola/
48     total
sachingola@DESKTOP-S4K7IU0:~$ cat > manipulate.txt
hello rohit
hello kohli
hello root
hello smith
^C
sachingola@DESKTOP-S4K7IU0:~$ diff duplicate.txt manipulate.txt
1c1
< hello kohli
---
> hello rohit
5,7d4
< hello root
< hello root
< HELLO ROOT
sachingola@DESKTOP-S4K7IU0:~$ diff -u duplicate.txt manipulate.txt
--- duplicate.txt      2024-01-28 20:16:53.115404773 +0530
+++ manipulate.txt     2024-01-28 22:14:02.580503783 +0530
@@ -1,7 +1,4 @@
-hello kohli
+hello rohit
 hello kohli
 hello root
 hello smith
-hello root
-hello root
-HELLO ROOT
```

```
sachingola@DESKTOP-S4K7IL  ×  +  ∨  -  □  ×
sachingola@DESKTOP-S4K7IU0:~$ history
 1  who
 2  whoami
 3  man sudo_root
 4  cal
 5  sudo apt update
 6  cal
 7  sudo apt install ncal
 8  cal
 9  who
10  w
11  whoami
12  pwd
13  whoami
14  date
15  cal
16  mkdir firstdir
17  cd firstdir
18  cat file1.txt
19  cat > file1.txt
20  cat >> file1.txt
21  cat file1.txt
22  head file1.txt
23  head -2 file1.txt
24  tail file1.txt
25  tail -2 file1.txt
26  cat > file2.txt
27  more file2.txt
28  less file2.txt
29  cd ..
30  cat > firstfile.txt
31  cat firstfile.txt
32  cat >> firstfile.txt
33  cat firstfile.txt
34  ls
35  pwd
36  cd firstdir
37  cp file1.txt /home/sachingola/firstfie.txt/
38  cp file1.txt /home/sachingola/
39  touch file2.txt
40*
41  cp file1.txt /home/sachingola/firstfile
42  cd ..
43  cat firstfile
44  cat > secondfile.txt
45  mv secondfile.txt /home/sachingola/firstdir/
46  cat seconfile.txt
47  cd firstdir
48  cat seconfile.txt
```

```
49 cat secondfile.txt
50 echo My Own Dell
51 $name=Sachin
52 $name='Sachin'
53 $name 'Sachin'
54 $name
55 echo name
56 echo $name
57 man echo
58 whoami
59 cat > file3.txt
60 cp file1.txt file3.txt
61 cat file3.txt
62 cd ..
63 mkdir seconddir thirddir fourthdir
64 ls
65 rm firstfile.txt
66 cat firstfile.txt
67 rmdir fourthdir
68 ls
69 rmdir firstdir
70 rmdir -r firstdir
71 man rmdir
72 rmdir -d firstdir
73 rmdir --help
74 rmdir -r firstdir
75 cd firstdir
76 rmdir -r firstdir
77 cd ..
78 rm -r firstdir
79 ls
80 cat > thirdfile.txt
81 cut -2 -6 thirdfile.txt
82 cut -2-6 thirdfile.txt
83 man cut
84 cut -c 1-6 thirdfile.txt
85 cut -d ' ' -f 2 thirdfile.txt
86 cat < anum.txt
87 cat > anum.txt
88 cat < anum.txt
89 rm a*
90 cat < anum.txt
91 cat > duplicate.txt
92 uniq -c duplicate.txt
93 uniq -d duplicate.txt
94 uniq -u duplicate.txt
95 cat >> duplicate.txt
96 uniq -c duplicate.txt
97 uniq -i duplicate.txt
```

```
sachingola@DESKTOP-S4K7IIL × + v
93 uniq -d duplicate.txt
94 uniq -u duplicate.txt
95 cat >> duplicate.txt
96 uniq -c duplicate.txt
97 uniq -i duplicate.txt
98 uniq -i -c duplicate.txt
99 ls -l
100 chmod u-w duplicate.txt
101 ls -l
102 chmod 644 duplicate.txt
103 ls -l
104 chmod u-r duplicate.txt
105 cat duplicate.txt
106 chmod u+r duplicate.txt
107 cat duplicate.txt
108 man chmod
109 chmod u=r duplicate.txt
110 ls -l
111 du duplicate.txt
112 du /home/sachingola/seconddir
113 du -h /home/sachingola/seconddir
114 du -s /home/sachingola/seconddir
115 du -c /home/sachingola/
116 cat > manipulate.txt
117 diff duplicate.txt manipulate.txt
118 diff -u duplicate.txt manipulate.txt
119 cat >> manipulate.txt
120 diff duplicate.txt manipulate.txt
121 cat >> manipulate.txt
122 diff duplicate.txt manipulate.txt
123 history
sachingola@DESKTOP-S4K7IU0:~$ !122
diff duplicate.txt manipulate.txt
1c1
< hello kohli
---
> hello rohit
7a8
> Hello Bro
sachingola@DESKTOP-S4K7IU0:~$ last

wtmp begins Sun Jan 28 16:43:57 2024
sachingola@DESKTOP-S4K7IU0:~$ man last
sachingola@DESKTOP-S4K7IU0:~$ last -a

wtmp begins Sun Jan 28 16:43:57 2024
sachingola@DESKTOP-S4K7IU0:~$ last -d

wtmp begins Sun Jan 28 16:43:57 2024
```

```
sachingola@DESKTOP-S4K7IL × + ▾  
sachingola@DESKTOP-S4K7IU0:~$ sort duplicate.txt  
HELLO ROOT  
hello kohli  
hello kohli  
hello root  
hello root  
hello root  
hello smith  
sachingola@DESKTOP-S4K7IU0:~$ sort -u duplicate.txt  
HELLO ROOT  
hello kohli  
hello root  
hello smith  
sachingola@DESKTOP-S4K7IU0:~$ cat duplicate.txt  
hello kohli  
hello kohli  
hello root  
hello smith  
hello root  
hello root  
HELLO ROOT  
sachingola@DESKTOP-S4K7IU0:~$ sed 's/root/morgan/g' duplicate.txt  
hello kohli  
hello kohli  
hello morgan  
hello smith  
hello morgan  
hello morgan  
HELLO ROOT
```



```
sachingola@DESKTOP-S4K7IU0:~$ sed '3i\Rohit is a good captain' duplicate.txt
hello kohli
hello kohli
Rohit is a good captain
hello root
hello smith
hello root
hello root
HELLO ROOT
sachingola@DESKTOP-S4K7IU0:~$ cat duplicate.txt
hello kohli
hello kohli
hello root
hello smith
hello root
hello root
HELLO ROOT
sachingola@DESKTOP-S4K7IU0:~$ grep "root" duplicate.txt
hello root
hello root
hello root
sachingola@DESKTOP-S4K7IU0:~$ grep "root" duplicate.txt manipulate.txt
duplicate.txt:hello root
duplicate.txt:hello root
duplicate.txt:hello root
manipulate.txt:hello root
manipulate.txt:hello root
manipulate.txt:hello root
sachingola@DESKTOP-S4K7IU0:~$ grep -i "root" duplicate.txt manipulate.txt
duplicate.txt:hello root
duplicate.txt:hello root
duplicate.txt:hello root
duplicate.txt:HELLO ROOT
manipulate.txt:hello root
manipulate.txt:hello root
manipulate.txt:hello root
manipulate.txt:HELLO ROOT
sachingola@DESKTOP-S4K7IU0:~$ grep -e "root" -e "kohli" duplicate.txt
hello kohli
hello kohli
hello root
hello root
hello root
```

```
sachingola@DESKTOP-S4K7IU0:~$ grep -r "root" /home/sachingola/
/home/sachingola/duplicate.txt:hello root
/home/sachingola/duplicate.txt:hello root
/home/sachingola/duplicate.txt:hello root
/home/sachingola/.bash_history:man sudo_root
/home/sachingola/manipulate.txt:hello root
/home/sachingola/manipulate.txt:hello root
/home/sachingola/manipulate.txt:hello root
/home/sachingola/.bashrc:# set variable identifying the chroot you work in (used in the prompt below)
/home/sachingola/.bashrc:if [ -z "${debian_chroot:-}" ] && [ -r /etc/debian_chroot ]; then
/home/sachingola/.bashrc:    debian_chroot=$(cat /etc/debian_chroot)
/home/sachingola/.bashrc:    PS1='${debian_chroot:+($debian_chroot)}\[\033[01;32m\]\u@\h\[\033[00m\]:\
[\033[01;34m\]\w\[\033[00m\]\$ '
/home/sachingola/.bashrc:    PS1='${debian_chroot:+($debian_chroot)}\u@\h:\w\$ '
/home/sachingola/.bashrc:    PS1="\[e]0;${debian_chroot:+($debian_chroot)}\u@\h: \w/a\]$PS1"
sachingola@DESKTOP-S4K7IU0:~$ grep -r "kohli" /home/sachingola/
/home/sachingola/duplicate.txt:hello kohli
/home/sachingola/duplicate.txt:hello kohli
/home/sachingola/manipulate.txt:hello kohli
sachingola@DESKTOP-S4K7IU0:~$ ls | grep "manipulate"
manipulate.txt
```

